

Water management and electricity output of a Hybrid Renewable Energy System (HRES) in Fournoi Island in Aegean Sea

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ABSTRACT

The necessity of mitigating the consequences of energy dependence and water scarcity in remote areas, such as small islands puts forward the investigation of renewable energy sources. Especially in Aegean, the rich wind potential and natural topography favors the use of Hybrid Renewable Energy System (HRES). In this research work, a HRES in Fournoi island, in eastern Aegean sea, is designed in order to utilize hydropower for electricity generation and cover drinking and agricultural water demands through desalination of sea water. The proposed system includes four wind turbines, a desalination plant, a small hydroelectric station, a pumping station and two water reservoirs. Concerning the generated power from the wind turbines, it is assumed that 30% is incorporated directly to the power grid, while the remaining 70% is available for water pumping and desalination. In order to evaluate the performance of the system, three scenarios are being studied, based on different combinations of fulfillment of electricity and water needs and a ten-year reliability is conducted. This study leads to results of the project's potential in covering both water and electricity demands of the local society.

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1. Introduction

The rising penetration of renewable energy sources (RES) in the energy balance globally [1,2] and the institutional framework established over the last decades evidence that renewables are a key priority for the European Union (EU). Both the EU and individual states propose new policies regarding the use of renewables. The beginning is placed in 1997 with the Green Paper “Energy for the Future” [3] for the protection of the environment. Today the report of the National Action Plan for achieving the contribution of renewable energy to 20% in final energy consumption by 2020, results by Directive 2009/28/EC. It includes estimates for the development of the energy sector and the penetration of RES technologies by 2020. The key objectives of the strategy are to reduce equivalent CO₂ emissions by 20%, to increase the share of renewable energy to 20%, and to achieve energy savings of 20% or more. Greece, as a member of the EU, participates in the overall effort of the EU to increase the contribution of renewables in the

energy balance. The objectives of Greece for 2020, according to its current legal framework [4], are the increase of the share of RES by 20% in final energy consumption and by 40% in electricity consumption [5]. According to current estimations of Regulatory Authority for Energy (RAE), large hydroelectric systems and the use of other renewable energy sources apart from wind are not expected to contribute more than 10%. The availability of the primary sources and the low cost of exploitation in relation to other RES, shows that wind energy is the most significant contributor in order to achieve national targets [6].

In a large number of Mediterranean islands, both residents and tourists, cover their energy requirements, by either local autonomous power stations [7], which produce energy by consuming fossil fuels or the submarine connection to the mainland's national network of Public Power Corporation (PPC). In consequence, they are suffering from high-cost electricity generation, blackouts during high energy demands periods and oil products dependence for the local power stations and thus the rich potential of Renewable Energy Sources should be exploited in order to achieve their energy independency. In this context, a research of implementation of Hybrid Renewable Energy Systems (HRES_s) is developed. HRES_s produce energy based on the combination of different forms of RES

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such as wind and hydroelectric and on the excess energy of the storage using the Pumped-Storage Hydroelectricity method (PSH), a method totally friendly to environment and easily installed in areas where topography and water availability permits [8]. Today, over 200 systems using PSH method are installed with capacity over 129 GW [9]. Numerous studies have been published, where a solution which includes energy storage is implemented [10–24]. In Ref. [13], optimum sizing of a pumped storage plant in Crete is presented, where the energy of the wind is consumed from the pumps, while firm power is provided from hydro turbines in specified time periods of a day. Also, many studies concerning HRES are published focusing on the design and analysis [25–28] and on the optimization [29,30] of such a system. Some of the publications refer to small communities like Binalood region in Iran [31], while others focus on islands like Ireland [32] or smaller islands like Lencois in Brazil [33], Utsira in Norway [34], Samso in Denmark [35], a small island in Hong Kong [36], Karpathos [37] and Mykonos [38] in Greece. Also, in Greece the HRES of Icaria island has been constructed and feasibility studies has been taken place, which result that the plant is expected to produce 23 GWh/year and one third of that is coming from PSH energy [13]. Also, this kind of investment is attractive not only by considering strictly financial terms, but also calculating environmental benefits. Even the problem of the environmental impact of large-scale technical projects, such as a dam, which is a prerequisite of a hydroelectric project, can turn into an advantage through a well-designed study. A typical case is the Plastira Lake, in Karditsa, in Greece, where the deluge of the region by waters after the creation of the dam, has created a new wetland, which soon turned into a tourist attraction and contributed to the evolution of ecotourism in the region.

Water scarcity for domestic and agricultural purposes is another problem that many Greek islands have to deal with. Tankers transfer water from the mainland, a particularly costly process and households have private rainwater harvesting tanks for domestic use. To meet the need of drinking water, bottled water usage is increased, which also increases the cost of living. Also, groundwater quality and quantity problems appear, due to numerous and uncontrolled private wells for irrigation needs. Under these circumstances research turns to the direction of desalination. There is a large number of techniques in order to achieve desalination of seawater and they are classified, according to the energy type applied, into two categories [39]. The first one is thermal driven, while the second is electrically driven. Large desalination plants use conventional forms of energy and they have a significant cost that affects the cost of produced water. The desalinated water should be consumed near the area produced because the transfer, especially in long distances or high altitudes from the place of production, increases significantly the cost. A desalination plant can also use waste energy for its operation. Shih and Shih [40] show that heat quantities discharged from chemical and other industries to the environment as cooling water or in the air as cooling air can be used, under specific circumstances, in desalination, because the content of thermal energy is low, but the quantities are large.

Over the last decades, studies [41–45] over desalination systems utilizing renewable energy sources have been published. To select the appropriate combination of RES and desalination technology, a great number of factors should be taken into consideration [42], such as the size of the plant, the salinity of the feed water, the remoteness of the project, the availability of grid electricity, the technical infrastructure and the type and potential of the local renewable energy resource and of course energy cost of the desalination technology. The research in the desalination industry has focused on the combination of desalination plants with RES, in order to reduce energy costs and therefore the cost of the produced desalinated water [46]. In Ref. [47] the feasibility of operating the

process of desalination using solar power is presented. Results show efficiencies that range from 65% to 90% and make the process sustainable. In Ref. [48] a case study of desalination of brackish groundwater using wind-generated electricity is implemented, with profitable results compared to high prices of electricity. Among all desalination processes, reverse osmosis (RO) finds its greatest application, because it can reach very high efficiencies and in combination with RES, low installation and operation cost and low CO₂ emissions [49] can be achieved. Energy consumption of RO desalination process range from 2.5 kWh/m³ to 7 kWh/m³ [50]. The fact that the global capacity of desalination plants is constantly increasing, makes easy to understand the importance of desalination to the problem of water scarcity in some countries, and this means that in the years to come it may be one of the most secure ways to tackle this problem.

In this research work an evaluation of a Hybrid Renewable Energy System (HRES) is developed in order to utilize hydropower for electricity generation and cover domestic water and irrigation demands through the operation of a reverse osmosis (RO) desalination plant. A methodological framework is developed for Mediterranean islands, using meteorological and topographical data. These islands very often suffer from lack of energy and water autonomy, although they possess proper wind conditions and geospatial characteristics. So, this proposed framework can be adapted easily for other isolated islands, using their own geographical, meteorological, population and demand (water, energy) data. The results give valuable information about the HRESs and the controlled distribution of the generated clean energy for electricity and water demands. This process leads to the conclusions about the functioning of the system and the reliability of this project.

2. Study area and data

Fourni (or Fourni Icarias/Korseon), is a cluster of islands located southwestern of Samos and eastern of Icaria, in the eastern Aegean Sea (Fig. 1). Island's coastline is about 120 km, exceeding in length this of the nearby Samos, which is an island ten times greater in surface. The inhabited islands of the cluster are: Fourni, Thimena and Aghios Minas. Thimena's surface is 10 km² with 151 inhabitants in the two settlements (Thimena and Keramidou) and Aghios Minas has 2.1 km² surface and only 3 inhabitants. Fourni is the largest among them with an area of 30.5 km² and it is also the capital city with 1400 inhabitants. However, tourism is a major

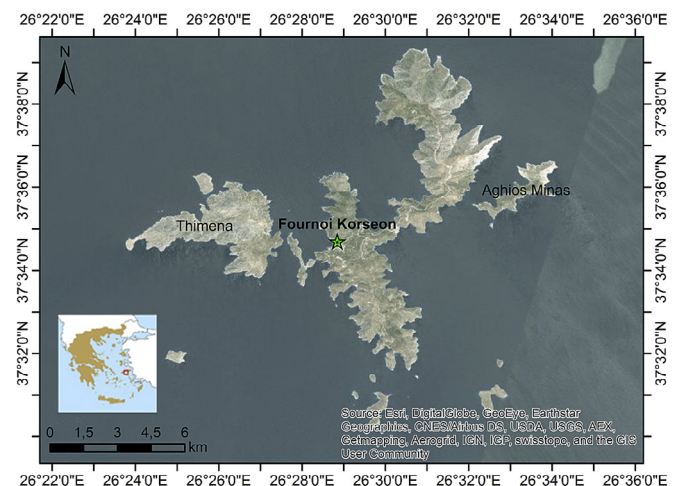


Fig. 1. Study area (Fourni island).

factor and on summer population reaches the 4000. It has its own municipality, Fourni Korseon, and it belongs to Samos Prefecture. According to the database for the Greek Nature Filotis [51] the entire island is a Natura habitat. A large number of monuments like the Acropolis, the Ancient Settlement of Kamari, the Ancient Quarry in Petrokopio and Sarcophagus, prove the long history of the island.

The climate in Fourni Korseon, due to its geographical position and the influence of the sea, is described as Mediterranean with maritime character, mild winters and long, warm and dry summers, low annual rainfall and plenty of sunshine throughout the year. According to Hellenic National Meteorological Service HNMS, the average annual temperature is about 18 °C, the monthly average minimum is 10 °C (in February) and the mean monthly maximum 28 °C (in July). The prevailing wind directions are mainly westerly and northerly by speed over 12.5 m/sec.

Although the greatest percentage of inhabitants lives in the settlement of Fourni, they do not have permanent access to drinking water. Households use private rainwater harvesting tanks, for domestic use and residents have access to drinking water, on a daily basis from 8:00 a.m. to 10:00 p.m., through a public tap, located in the central square of the island. The limited drinking water supply, has led to widespread use of bottled water. Concerning the agricultural use and according to data provided by Hellenic Statistical Authority, most of the island's area, namely 76%, is covered by scrub vegetation, in a much smaller proportion of 14.5% grazing land, areas with sparse vegetation (2.0%), land covered mainly by agriculture with significant areas of natural vegetation (1.2%), intermittent civil construction (1.1%), transitional wooded scrublands (1.1%) and indeterminate coverage (4%). There is no municipal water intake available for agricultural use and irrigation takes place through private wells. For its electricity needs, the island is submarine connected with the PPC via the nearby island of Samos. It should be noted that the island has comparative advantages in RES sector due to climatic conditions. So far, the wind potential is unexploited.

Data about the island's permanent population and data of water demands are provided by the Technical Service of the municipality of Fourni. Monthly variation of the island's demands in water for domestic use and irrigation is shown in Fig. 2a. The demand for water supply is increased during the summer months because of tourism and irrigation season. The minimum monthly demand reaches the value of 5400 m³ from October to April and the maximum value of 15600 m³ on August. Water for irrigation purposes is demanded from April to September, with the minimum monthly irrigation demand on April (6400 m³) and the maximum on July (19800 m³).

Data about island's electricity demands are provided by the PPC of Icaria, for 2015 and their monthly variation is shown in Fig. 2b. The demand for electricity is increased during the winter months, reaching the value of 665000 kWh, due to increased need for hot water and heating, energy consuming activities. Also, increased

demand is observed during the months of July and August due to the increase of residents through tourism. The minimum electricity demand of 163000 kWh is observed on September.

3. Methodology

The proposed HRES consists of two water reservoirs, one at sea level and a second at a higher altitude, four wind turbines 900 kW each, a desalination unit 500m³/day, a pumping station 300 kW, which uses water either directly from sea or after desalination and a hydroelectric station 5 m³/s. The turbine proposed to be used in this project is a reaction turbine and specifically a Francis model [52] with 90% efficiency. Upper and lower reservoir capacity is estimated using the conventional Rippl method [53] taking into consideration the inflows and outflows of each one. The upper reservoir is estimated to be 200000 m³, while the lower is estimated to be 340000 m³, as this reservoir supplies the desalination unit for the coverage of drinking and agricultural water demands. The water of the upper reservoir supplies the hydroelectric station, which covers the needs of electricity. If the demand for electricity cannot be covered by the generated energy, then the deficit is covered by the submarine connection of the island with the network of Samos. Detail of the island, in Fig. 3a, provided by the National Cadastre & Mapping Agency s. a., shows the location of the four wind turbines, the upper and the lower reservoirs and the archeological sites of the island. Due to limitations of legislation and spatial criteria [54], the system is installed in such a way as not to be visible from the four Archeological monuments of the island. Also, the schematic representation of the HRES, in Fig. 3, shows the way every individual unit of the project is connected to each other in the direction of utilization of hydropower. It is assumed that in all scenarios, the 30% of the total wind turbines energy production is sent directly in the local power network, while the remaining 70% is used for pumping and/or desalination depending on the scenario. In the first scenario (Fig. 3b), the 70% of the energy is used only for pumping water in order to exploit hydropower to meet the electricity needs and there is no desalination unit. In the second scenario (Fig. 3c), 70% of the energy is used for both pumping water to meet the electricity demands and desalination to meet domestic and irrigation demands. In the third scenario (Fig. 3d), only electricity needs are considered, as in scenario I, with the difference that desalinated water is used through the hydroelectric station.

Measurements of the wind speeds and temperatures in the weather station of HNMS, in Icaria Island, in 2015, are used in this study. Analyzing the direction of winds shows that this is mostly in the west and that is why the wind turbines are chosen to be placed in this direction. To calculate the required irrigation water according to the crops of the island, the Blaney-Criddle method is used.

Four wind turbines from Enercon, model 44–900 kW, are used. The power curve is provided by the company for this model, in order to calculate the wind power output from the wind turbine,

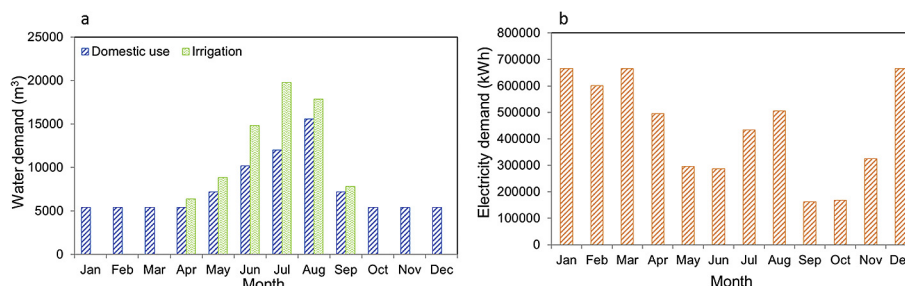


Fig. 2. Mean monthly demands: (a) Water for domestic use and irrigation, (b) Electricity.

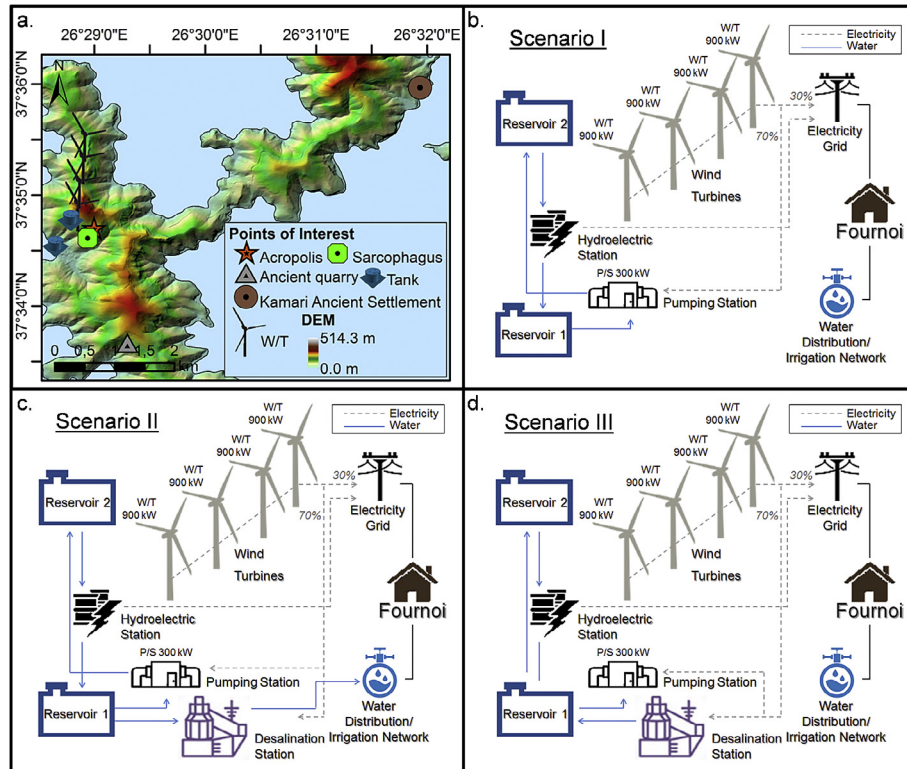


Fig. 3. (a) Detail of Island Fournoi, showing the location of the project and some points of interest, (b, c, d) Schematic representation of the HRES for scenarios I, II, III, respectively.

which depends on the wind speed at the hub height (u_2). According to the power curve, their maximum power (900 kW) is succeeded when wind speed u_2 is over 17 m/s. For wind speeds at the hub height lower than 2 m/s no power output is measured. The average annual wind speeds of the study ranging between 7 m/s to 9 m/s.

Wind turbines and weather station are placed in different heights. To derive the wind speed at the hub height from the speed at the weather station the following formula [2] is used:

$$u_2 / u_1 = \ln(z_2 / z_0) / \ln(z_1 / z_0), \quad (1)$$

where, u_1 : the wind speed at the weather station (m/s), u_2 : the wind speed at the hub height (m/s), z_1 : the weather station height (m), z_2 : the hub height (m), z_0 : the roughness length (m), which takes different values depending on the natural surface of the installation.

The function of the hydroelectric station is based on water that is pumped from the reservoir at sea level to the other at the higher altitude, when electricity demands are low. When electricity demands are higher the water from the upper reservoir is released and through the hydroelectric station produces energy for the electricity grid of the island as it heads on the lower reservoir where it is collected again for further use. The amount of generated energy is proportional to the altitude of the second reservoir and the volume of the available water. The water volume that is stored in the upper reservoir is added to the remaining water volume in the upper reservoir from the previous time step. Although more energy is required in order to pump the water to the higher altitude than the amount that is generated during the release of the water, however, the advantage of this operation is the ability to store excess energy and use it during high demand periods of the day, or low wind potential periods.

In order to control the response of the hybrid system and the

reliability rate to the required demand of electricity and water supply and irrigation, three scenarios are investigated, with time step of 24 h. Using formula (1) for the calculation of wind speed at the hub height, and using the output power curve of the company of the wind turbine, the wind power output is calculated for each day. In all scenarios, it is assumed that 30% of the total wind turbines energy production is sent directly in the local power network for electricity needs, while the remaining 70%, is used for the water pumping to the higher reservoir and desalination, depending on the scenario tested. The ratio of 30/70% is selected due to the stochastic nature of the wind, and it is based on the common practice used in Greece. Furthermore, the selected energy allocation ratio in this work is also confirmed through an analysis of different ratio values within a range of 10/90% to 40/60%, which concludes that the best performance of the proposed system occurs when the ratio is 30/70%. Although wind energy is a low-energy technology that is not affected by unpredictable and unstable fossil fuel prices, it depends on a considerable extent on the availability of wind resources and, due to the inherent stochastic nature of the wind, is highly fluctuating in electricity generation energy. As a result, the larger proportion of the produced wind energy is preferred to be used for energy storage in order to cover high energy demand periods. Fig. 4 stands for the illustrated algorithm, which contains the methodology of each scenario and its description is given in the following paragraphs.

The first scenario is tested to meet exclusively the demand for electricity in the island and that means that all the amount of generated energy is used for the electricity needs, without giving any energy in the desalination unit for the needs of drinking water and irrigation. In this scenario 70% of the total wind turbines energy production is used for pumping water from the lower reservoir to that at the higher altitude, so as energy to be produced by hydroelectric station for electricity needs. The available volume of water,

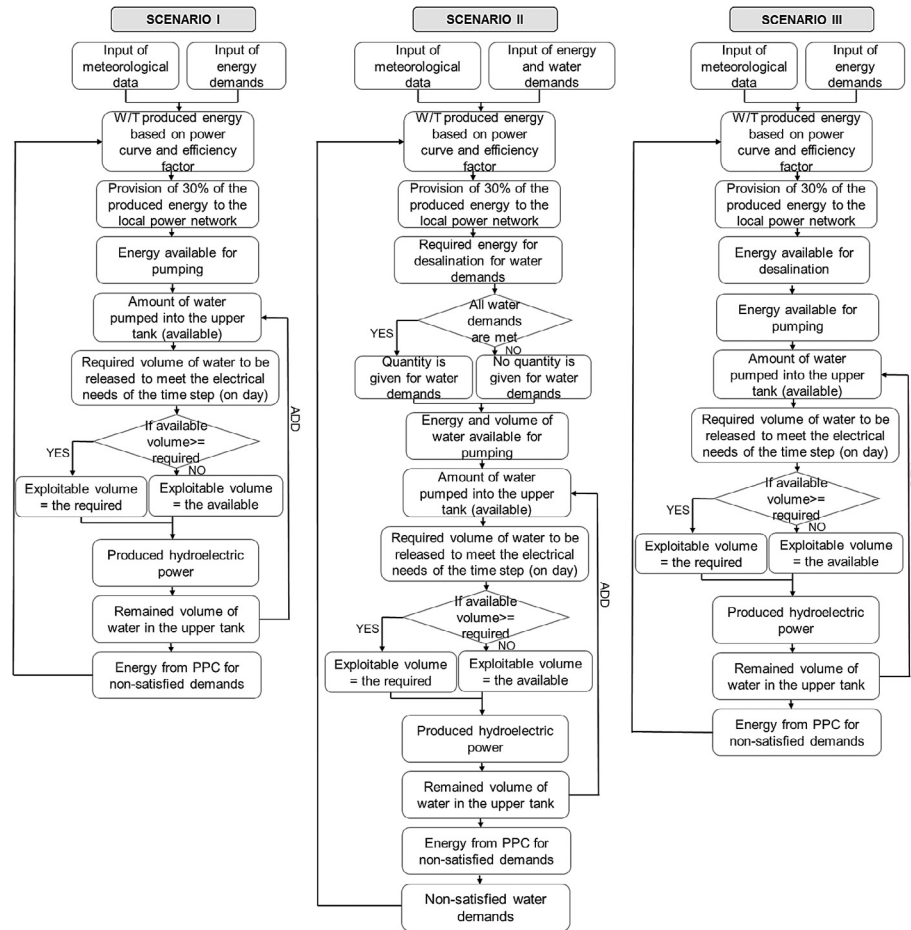


Fig. 4. Flowchart of the methodological framework (scenario I, II, III).

that can be pumped to the upper reservoir, is calculated and is added each time to the available volume from the previous step. After that the produced energy from the hydroelectric station is calculated and a check is taken place to ascertain whether the electricity demands are covered for this time step. Demand that is not covered, is supposed to be provided by the PPC and is also calculated for each time step. When both the hydroelectric station and the 30% of the wind turbines power cover the total electricity demand, then the island does not use at all the PPC via the submarine connection with the nearby island of Samos.

The second scenario is considered to meet both the demand of electricity and the needs of desalinated water for domestic use and irrigation. In this case the 70% of the total wind turbines energy is used for desalination of seawater. Priority for one day (time step) is given to fulfill the water demands. The energy, which is required for desalinating the amount of water demand is calculated. Also, it is checked if the demand for 24-h can be fully covered. Otherwise, it is supposed that the uncovered amount of water, will be satisfied by tankers. The excess produced energy from the hydroelectric station is used for pumping water from the lower reservoir to that at the higher altitude in order to produce, energy for high demand periods. Same as in scenario I, depending on the excess energy the available volume of water, that can be pumped to the upper reservoir, is calculated and is added each time to the available volume from the previous step. The produced energy from the hydroelectric station is calculated and it is examined whether the electricity demands are covered for this time step. When both the hydroelectric station and the 30% of the wind turbines power

cannot cover the total electricity demand, then the island use the submarine connection with the PPC.

The third, and last, scenario that is examined is considered to meet only the demand of electricity needs, same as at the first scenario, but it requires the desalination of the total quantity of water that goes through the hydroelectric station, although no amount is intended to be used for domestic or agricultural use. This procedure contributes to the avoidance of deterioration of the pump from the salts of the seawater. In this case, also a 30% of the total wind turbines energy production is sent directly in the local power network for electricity needs, while the remaining 70% is used firstly for desalination of an amount of water and then for pumping this amount, in order energy to be produced by hydroelectric station for the electricity needs. For each day, based on the simulation procedure, the amount of water desalinated and pumped is calculated, depending on this excess 70% of the produced energy. According to the final desalinated amount of water in the upper reservoir, the produced energy from the hydroelectric station is determined and it is examined whether the electricity demands are being covered, otherwise the deficit is covered by the PPC.

4. Results

4.1. Scenario I

The contribution power of each energy source (wind turbines, hydropower and energy from the PPC network) to cover the entire island's power demand, is shown in Fig. 5a. The total contribution

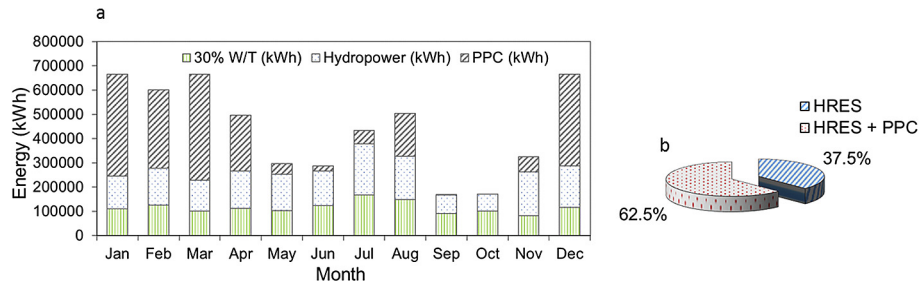


Fig. 5. Scenario I: (a) Monthly coverage of electricity demand by three energy sources, (b) Annual reliability rate for the electricity coverage.

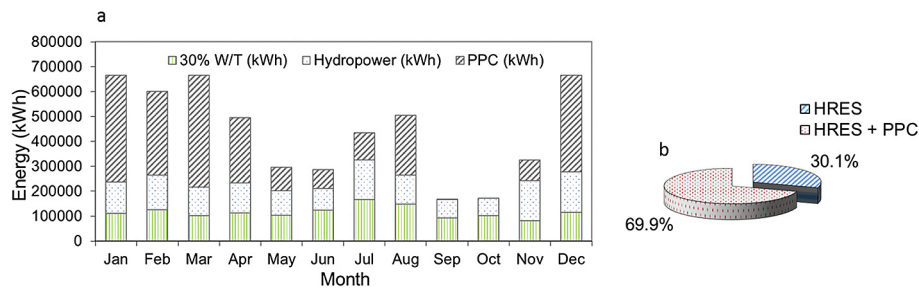


Fig. 6. Scenario II: (a) Monthly coverage of electricity demand by three energy sources, (b) Annual reliability rate for the electricity coverage.

of the national network of PPC during the year is greater than the contribution of the wind turbines and the contribution of the hydropower, but there are months where the quantity demanded is mainly covered by these two sources by far. Also, it is observed that especially in the peak period, from May to August, high percentage coverage is achieved from the HRES. Furthermore, during the months of September and October the full autonomy of the island is achieved.

The first scenario, results in the reliability of the system, which means the number of days, during a year, when the island manages to be energy independent because of the hybrid system and is shown in Fig. 5b. In a percentage of 62.5% of the total year, the island cannot meet the needs of electricity through the hybrid system, while the remaining 37.5% of the year the island can be autonomous and there is no need for connection to PPC.

4.2. Scenario II

The second scenario, which is considered to meet demand for electricity and the needs of desalinated water for domestic use and irrigation, shows the corresponding values of the contribution of each energy source and the reliability of the system in Fig. 6a and b respectively. In Fig. 6a a similar situation to the first is observed for the contribution of the three sources of energy. In the same way during the months of September and October the island is fully autonomous. Also, in the peak season, from May to August, which includes the summer and the irrigation period, relatively high percentage coverage is achieved from the HRES. In a percentage of 69.9% of the total year, the island cannot meet the total needs of electricity using only the hybrid system, while the remaining 30.1% of the year the island can be autonomous and there is no need to connect to the PPC. An increase of the PPC's contribution to final consumption compared to the first scenario is observed, something expected as in this scenario more energy is required in order to meet with drinking and irrigation water needs.

Fig. 7a shows the monthly amount of water supply, for domestic

use, provided by desalination (HRES) and the quantity that is not covered, while Fig. 7b shows the percentage of the annual water supply fulfillment. It should be noted that on a monthly basis the HRES can fulfill the domestic water demand within an annual range of 66–97%, while these percentages range between 87 and 94% in the peak period of tourism, which can be considered as an encouraging result for the water supply of the island. The 86.4% of the total year, the island can meet its domestic water supply needs through the HRES, while only in the 13.6% of the year the use of conventional water supply (public tap, water transfer, bottled water) is also required.

The monthly amount of irrigation water provided by desalination (HRES) and the respective quantity that is not covered is presented in Fig. 8a. Monthly percentages of irrigation water

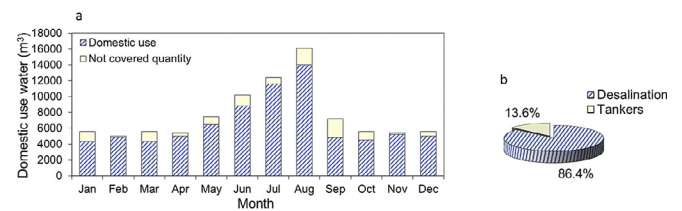


Fig. 7. Demand coverage of drinking water (scenario II): (a) monthly, (b) annually.

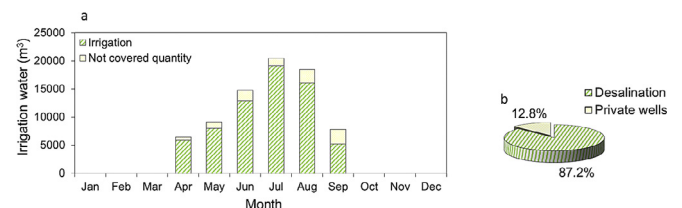


Fig. 8. Demand coverage for irrigation (scenario II): (a) monthly, (b) annually.

fulfillment through the HRES operation range between 66% and 94% and more specifically, the month with the peak demand for irrigation water (July), HRES covers the 94% of the total need. The annual performance of the HRES can be observed in Fig. 8b, where the percentages of the total irrigation needs fulfillment for the whole year are presented. Similarly, to the monthly results, in an 87.2% of the total year, the island can meet its irrigation needs through the HRES, while in the remaining 12.8% of the year, the use of private wells or municipal water for irrigation water need is required as a supplement. In this scenario, the generated energy sharing for electricity and desalination mainly accounts for the small quantities of water (domestic and irrigation) required to support the HRES operation.

4.3. Scenario III

By studying the third scenario, which is requiring the desalinating of the total quantity of water before taken to the hydroelectric station, the results show in Fig. 9a that the energy needs of the island during a year are covered by 26.2% from the direct use of 30% of the power of the wind turbine, by 17.3% from hydropower and the remaining 56.5% coming from the PPC, when the corresponding percentages without desalination are 26.2, 34.7 and 39 respectively. Energy from the wind turbines remains unchanged, as expected, but in this scenario more energy is needed in order to desalinate the entire quantity of water before taken to the hydroelectric station and this has as a result the need of more contribution of the PPC.

Ultimately, the reliability of the system, i.e. the percentage of days during a year that the island manages to be energy independent, because of the hybrid system, reaches, as it is shown in Fig. 9b, the rate of 21.1%. The corresponding rate without desalination is 37.5%.

4.4. Ten-year reliability

The study of the reliability of the project covering electricity demands, by the direct disposal of the 30% of the total wind

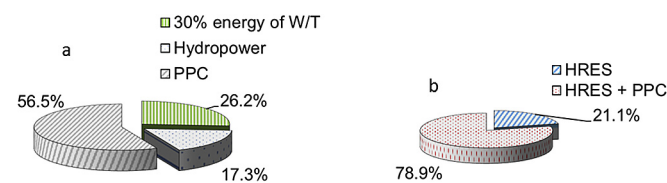


Fig. 9. Scenario III: (a) Annual coverage percentage of electricity demand by three sources, (b) Annual reliability rate for the electricity coverage.

turbines energy production and the generated energy from the hydroelectric station, is presented in Fig. 10. The methodology follows the same procedure as in scenario II, where the purpose is to cover both the electricity and the drinking and irrigation water needs. The results are focused on the ability of energy independence of the island for the fulfillment of electricity needs. It is assumed that the needs of the island (electricity and water) remain the same, but there is a variation in wind and temperature data, as they have been measured at the same weather station in Icaria Island and provided by HNMS for each year separately from 2005 since 2015. The results show a variation depending on the wind and temperature values prevailing in the area, as these are the two variables that are changing through these ten years. The lowest rate is 24.9% in 2008 and the highest at 38.6% in 2012. In 2015, the year for which the whole project is studied, gives the result of 30.1% as presenting also in Fig. 6a.

In Fig. 11, the monthly variation of the electricity demand coverage for a 10-year period is shown. The 30% of W/T, the hydropower generation and the PPC. Scenario II is used, where electricity and drinking and irrigation water needs are supposed to be covered. The results of the electricity fulfillment show that every year in October the island can meet its whole electricity needs and be autonomous. From the high values of wind turbine power that are observed for this month, it is assumed that the autonomy is a result of high wind potential. Very satisfactory results are observed in September too. On winter months, the use of the PPC is increased, due to the increased need in electricity. Significant is the fact that in July 2012 the island makes low use of the PPC, although it is a summer month with increased tourism. This suggests that, this year, higher winds are prevailing in the area, compared to the rest of the ten years tested, so the wind turbines are working at their maximum power and more energy is supplied to the grid and to the pumping and desalination units. On the other hand, in July 2011 extensive use of the PPC is observed, perhaps due to low winds, as the production of energy from wind turbines is quite low this month.

5. Conclusions and recommendations

In this research work, the development of a HRES in Fourni, a small island in northern Aegean, is investigated focusing on energy independency in remote areas. The proposed methodological framework is designed in order to produce hydropower for the coverage of electricity needs and for the desalination of the amount of water that is needed for domestic and agricultural use. Data about the island's population and data of water and electricity consumption are collected, as well as, meteorological data from the nearby station. Analysis of the consumptions data shows high

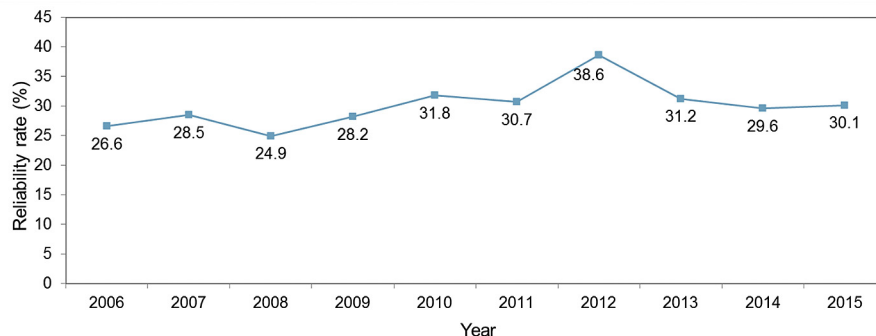


Fig. 10. Annual reliability rate for the electricity coverage in a 10-year period.

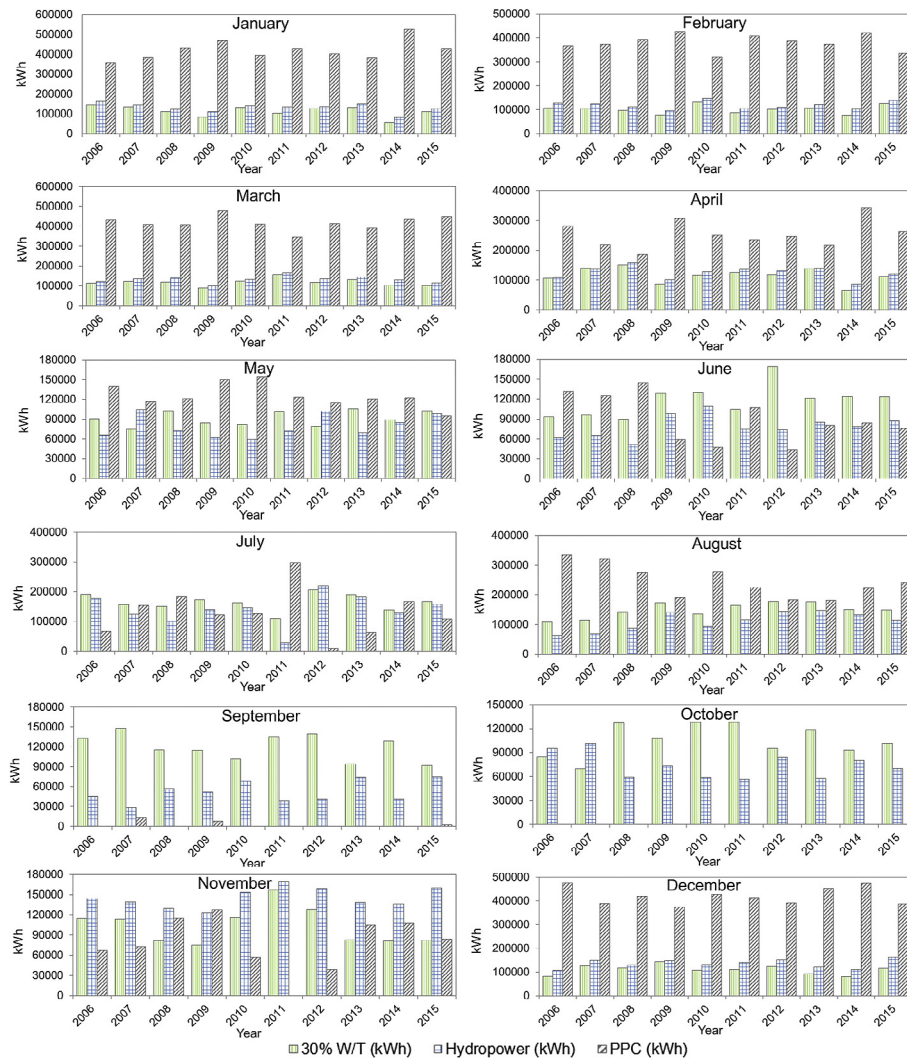


Fig. 11. Monthly coverage of electricity demand by three energy sources in a 10-year period.

values of water demand during summer period, because of tourism, and high values of electricity needs during the winter months, due to unfavorable weather conditions, resulting in operation of high energy consumption devices. Analysis of wind characteristics indicates the appropriate direction for the wind turbines installation in order to exploit the wind potential of the region. Three different scenarios are implied depending on possible different use of the suggested system, in order to investigate the performance of the proposed system at various operating conditions. The first scenario (scenario I) is to meet exclusively the demand for electricity in the island and the second one (scenario II) is to meet both the demand of electricity and the needs of desalinated water for domestic use and irrigation. The third scenario (scenario III) is same as scenario I (meets exclusively the electricity demand), with an additional requirement of desalinating the total quantity of water that goes through the hydroelectric station. It is found that scenario I provides more days of autonomy in electricity compared to the other two scenarios, as the amount of 70% of the produced energy is used only for pumping of water for hydropower. In scenario III the percentage of the days that the island is energy independent is decreased by 16.4% annually, since an amount of energy is consumed for desalination purposes. In scenario II that considers mainly the water supply, the HRES can cover more than 86% of annual water demand including the peak period of tourism and

irrigation. Also, although in this scenario electricity demand coverage is somewhat lower than in scenario I (30.1% comparing with the percentage of 37.5 for scenario I), HRES remains the principal source of energy during the period of May to August. The analysis of the ten-year reliability shows a slight variation, which occurs due to different wind and temperature data in the area during each year and reveals that even though the contribution of PPC is essential in annual scale, HRES has dominant role on the coverage of electricity needs in one third of the year (e.g. May to July and November) and there are months (e.g. September and October) of energy independency.

Based on the aforementioned encouraging results, further research is suggested in the direction of uncertainty analysis of the system input data. A stochastic generation of synthetic time series data would provide a more complete picture of the island's needs coverage and enhance more the reliability of the results.

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